

SEO Website Audit Toolkit

Feature & Implementation Requirements for the Website Pre-Launch Checker

Objective: Extend the existing Website Pre-Launch Checker into a practical SEO auditing system. The system should inspect a supplied website/page, identify SEO issues, explain why they matter, and provide actionable recommendations. The tool should clearly distinguish deterministic checks from inferences and should never claim that a page is fully SEO-optimized based on a single signal.

1. Meta Tag Analyzer

Purpose: Inspect the page's primary SEO and social metadata and flag missing, duplicated, empty, or poorly sized fields.

Checks: title tag; title length; meta description; description length; canonical URL; robots meta tag; viewport; language attribute; Open Graph tags; Twitter/X Card tags.

Example output: ■ Title missing ■■ Meta description too long ■ Canonical found ■ og:image missing.

2. Heading Structure Analyzer

Purpose: Analyze the semantic heading hierarchy and identify structural problems that can affect content understanding and accessibility.

Checks: H1 exists; multiple H1s; empty headings; excessively long headings; skipped heading levels; heading hierarchy. Display a visual H1 → H2 → H3 tree.

Example: H1 → H3 without an H2 should be reported as a warning, not automatically treated as a fatal SEO error.

3. Image SEO Analyzer

Purpose: Find image-related SEO and performance issues across the audited page.

Checks: missing ALT; empty ALT; excessively long ALT text; width/height attributes; image format; dimensions; file size; lazy loading; filename quality.

Useful report: list each image, its URL/name, file size, format, ALT status, and recommended action. Flag filenames such as IMG_2398.jpg and suggest descriptive naming such as blue-sapphire-ring.jpg.

4. URL SEO Analyzer

Checks: HTTPS; URL length; uppercase characters; spaces; special characters; excessive query parameters; underscores versus hyphens; trailing-slash consistency; readability and descriptiveness.

5. Canonical URL Checker

Purpose: Detect canonicalization problems and duplicate canonical declarations.

Checks: canonical exists; absolute URL; self-referencing canonical; HTTPS; redirect behavior; multiple canonical tags; canonical target availability.

6. Robots.txt Analyzer

Purpose: Inspect robots.txt and identify accidental crawling restrictions.

Checks: file exists; accessible status; User-agent directives; Disallow/Allow rules; important paths potentially blocked; sitemap reference.

Priority example: If /products/ is blocked, surface it as a high-priority issue because important content may be prevented from crawling.

7. XML Sitemap Analyzer

Purpose: Validate the site's XML sitemap and compare sitemap URLs against crawl/indexability signals.

Checks: sitemap discovery; HTTP status; XML validity; URL count; HTTPS usage; broken URLs; redirecting URLs; noindex URLs; last-modification dates; sitemap references from robots.txt.

Example: 342 URLs found → 329 indexable, 8 redirects, 5 broken.

8. Broken Link Checker

Purpose: Crawl links within the audit scope and identify broken or redirected destinations.

Checks: internal links; external links; 2xx success; 3xx redirects; 4xx/5xx failures; timeout/network errors. Show source page, destination URL, HTTP status, and redirect target when applicable.

Engineering requirement: use controlled concurrency, timeouts, retry limits, and a crawl limit so the tool does not behave like an uncontrolled crawler.

9. Internal Linking Analyzer

Purpose: evaluate internal-link distribution within the pages actually crawled.

Checks: pages with very few internal links; pages with no incoming internal links within crawl scope; anchor text; repeated anchor text; important pages receiving insufficient internal links.

Important limitation: call these findings 'within the audited crawl scope' rather than claiming a site-wide orphan-page determination unless the entire relevant site has been crawled.

10. Structured Data / Schema Analyzer

Purpose: detect and inspect structured data, especially JSON-LD.

Detect: Organization, LocalBusiness, Product, Article, BreadcrumbList, FAQPage, WebSite, Person, and other schema types where identifiable.

Checks: valid JSON-LD; schema type; important properties; obvious missing required/recommended properties; duplicate/conflicting markup. Clearly label this as a basic analyzer unless a dedicated schema validation service is used.

11. Open Graph & Social Preview Analyzer

Checks: og:title, og:description, og:image, og:url, og:type, twitter:card, twitter:title, twitter:description, twitter:image.

UI requirement: generate a visual social-preview card showing approximately how the page metadata could appear when shared. This turns a technical check into a useful client-facing feature.

12. Mobile SEO Checker

Purpose: identify mobile-readiness signals and, where possible, verify behavior using a real browser render.

Static checks: viewport meta tag; obvious fixed-width constructs; responsive CSS indicators. **Rendered checks:** horizontal overflow, viewport behavior, visible layout problems, and other measurable signals.

Rule: do not label a page 'mobile-friendly' solely because a viewport tag exists.

13. Page Performance & Core Web Vitals Analyzer

Purpose: measure real browser performance rather than relying only on HTML inspection.

Metrics: LCP, CLS, INP, TTFB, DOM size, total page weight, request count, JavaScript size, CSS size, and image weight where measurable.

Output: metric value, threshold/status, likely cause, and actionable recommendation. Prefer a real browser automation approach for rendered measurements.

14. HTTPS & Security Header Checker

Purpose: add security-related pre-launch checks that complement SEO auditing.

Checks: HTTPS; HTTP→HTTPS redirect; HSTS; Content-Security-Policy; X-Content-Type-Options; Referrer-Policy; Permissions-Policy.

Note: this is a website quality/security module, not a replacement for a professional security assessment.

15. Indexability Checker

Purpose: answer the practical question: 'Does this page appear indexable based on the signals the tool can observe?'

Signals: robots.txt; noindex; X-Robots-Tag; canonical; HTTP status; redirects; page accessibility.

Output: Indexable / Not indexable / Uncertain, followed by the exact reasons. Avoid pretending that the tool can guarantee search-engine indexing.

16. Overall SEO Audit & Reporting

Dashboard: provide a single overall score plus category scores such as Technical SEO, On-Page SEO, Performance, Images, Social, Indexability, and Security.

Issue levels: Critical, Warning, Passed. Each issue should contain: what was detected, why it matters, how to fix it, affected URL(s), and evidence from the audit.

Recommended summary: Overall Score → Critical Issues → Warnings → Passed Checks → Detailed Modules → Recommendations.

Recommended Implementation Order

Phase	Modules	Priority
1 — Core SEO	Meta Tags; Headings; Images; URLs; Canonical; Indexability	Must have
2 — Technical SEO	Robots.txt; Sitemap; Broken Links; Internal Links; Schema	Must have

Phase	Modules	Priority
3 — Performance & Social	Open Graph; Mobile; Core Web Vitals; Security Headers	High
4 — Reporting	Scoring; severity system; recommendations; exportable report	High

Minimum Quality Requirements

1. Responsive UI for desktop and mobile.
2. Clear loading, error, empty, and timeout states.
3. Validate user-supplied URLs and restrict unsafe/uncontrolled crawling behavior.
4. Use sensible request timeouts, concurrency limits, crawl depth/page limits, and retry rules.
5. Never expose server-side secrets or API credentials to the browser.
6. Show evidence for every finding instead of producing unexplained scores.
7. Separate 'measured', 'detected', and 'recommended' information.
8. Make every audit result understandable to a non-technical client, while retaining technical details for developers.

Suggested Final Product

The finished system should feel like a lightweight internal SEO auditing product rather than a collection of random checkboxes. A user enters a URL, starts an audit, watches progress, receives a score and prioritized issues, opens detailed findings, and can export/share the final report.